Figure 5.1 An example of a disjoint-set. The whole set, the so-called Universe, is the set of "food". Three partitions are shown: fruit, vegetables, and sweets. The key point is that these subsets have no intersections with each-other.

We will tackle this problem starting with its definition, then cover the most basic (and naïve) algorithms, to give readers an understanding of what an actual solution looks like; once that's clear, we will be ready to delve into making our solutions more efficient, and show how to use it as part of more complex algorithms, so that by the end of the chapter you will be able to code up the best possible solution for disjoint sets, and employ it to improve the performance of higher level applications.

5.1 The Distinct Subsets Problem

For instance, imagine this: you are running a new, recently-created, e-commerce website, and for your launch, you'd like to provide non-personalized recommendations to your users. If it helps you feeling this like a real situation, you might imagine owning a time-machine and being back to 1999 – or more realistically, you can think about opening a geographically localized website with stronger ties to your Country's retailers, or maybe a specialized retail website, focusing on niche products. Either way, this can be a very interesting exercise.